Improving Metadata Compliance and Interoperability at EGA with Biovalidator

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Introduction

The **European Genome-phenome Archive** (EGA), an ELIXIR core data resource for archiving and sharing personally identifiable genetic and phenotypic data, continually strives for improved metadata compliance and interoperability in line with the FAIR principles. To achieve this, the EGA will: (1) apply its enhanced **metadata model** with the help of the JSON Schema definition; and (2) integrate it with **ELIXIR Biovalidator**, a comprehensive JSON Schema validator tailored for life sciences.

This poster illustrates how the EGA's enhanced metadata model, combined with the power of Biovalidator, fosters a central, open-sourced and remote metadata validation. This synergy not only bolsters the metadata FAIRness but also greatly benefits the Federated EGA (FEGA) initiative, by providing to emerging nodes an out-of-the-box workflow for metadata validation. The EGA aims to establish a process for constant review and incorporation of feedback into its model through collaboration with ELIXIR Human Data Communities.

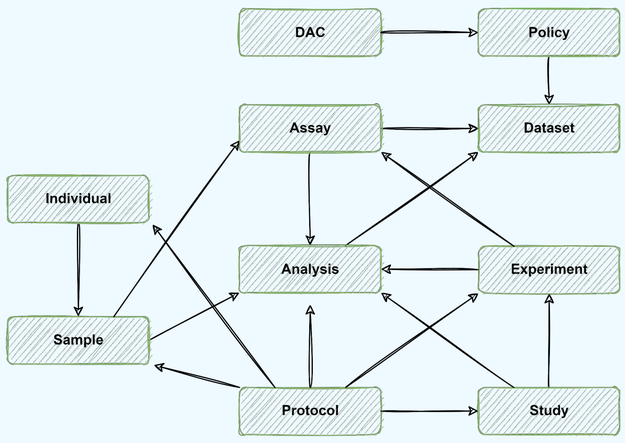
EGA Metadata schemas

In order to improve its metadata standards, the EGA has worked towards upgrading its existing metadata definitions. The outcome is a revised **set of JSON Schemas** hosted in GitHub.

EbiEga/ega-metadata-schema

still in development, collaboration is welcomed!

The schemas are organised in **modular files** that can be referenced. Whenever a validation request contains a reference to them, Biovalidator fetches and compiles them, if they are not already in cache.



Thanks to Biovalidator, we can encode **ontology based constraints**:

UBERON:0000465 — *Check term in UBERON ontology* → uberon

axilla skin

ELIXIR Biovalidator

The ELIXIR Biovalidator[1] is an advanced **JSON Schema validator** based on Ajv[2]. Biovalidator is publicly available at ELIXIR's GitHub repository.

elixir-europe/biovalidator

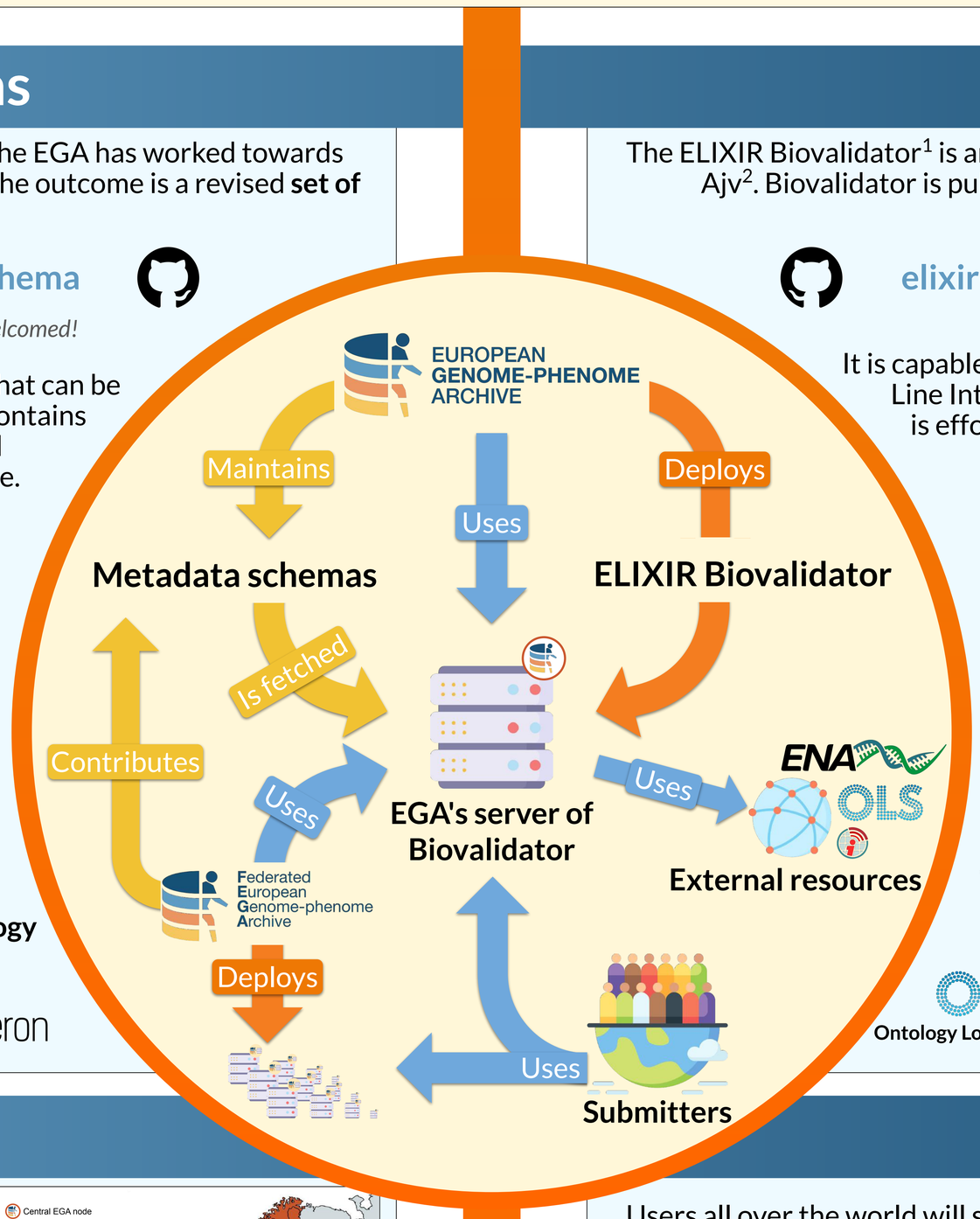
It is capable of running both as a **server** or in a Command Line Interface (**CLI**) mode. Its deployment as a server is effortless and, if the server port is exposed to the world, it allows for **anyone** to use it remotely, **anywhere** and **anytime**.

At EGA, we deployed our own server:

biovalidator.ega.ebi.ac.uk

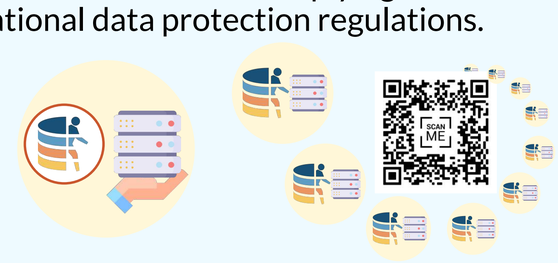
Its validation rules are augmented through the usage of **user-defined keywords**. These allow for semantic validation of the metadata relying on external resources, easing the maintenance of the metadata definitions

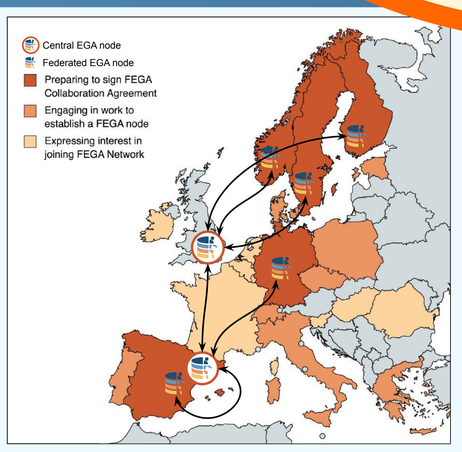
OLS Ontology Lookup Service[3] · identifiers.org[4] · ENA European Nucleotide Archive[5]



FEGA ecosystem

The **Federated EGA** (FEGA) is a network of EGA nodes across different European countries, working collectively for secure human data sharing and research advancement while complying with national data protection regulations.





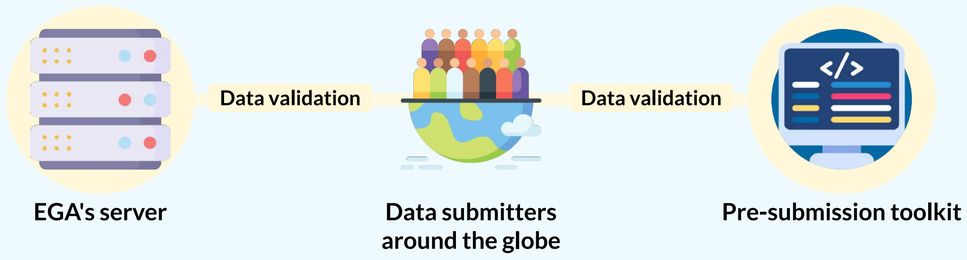
When metadata is validated through Biovalidator, the information itself is not saved at the EGA. Therefore, FEGA nodes can **utilize the existing EGA Biovalidator server** for metadata validation or choose **to deploy their own** local servers for more personalized control.

The true beauty of this project is that as long as the same schemas are referenced and the same version of Biovalidator is used, the **validation outcome will be the same**. This boosts interoperability and enables seamless synchronization between nodes.

Same outcome
- Same schemas
- Same deployment

Global community

Users all over the world will soon be able to easily **validate their metadata against EGA's standards.** This will be possible by either reaching our Biovalidator endpoint directly, or through a pre-submission toolkit that is in the making and being designed for lab-coat scientists and physicians.



The global community is invited to **contribute** to the development and enhancement of EGA's metadata schemas and ELIXIR's Biovalidator, driven by the needs of the worldwide user base.

By making metadata validation accessible, EGA aims to **bridge the gap between different data standards** across the world, facilitating more comprehensive and meaningful data analysis and comparisons.

Contact information

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- More about **Federated EGA**: Mallory Freeberg (mfreeberg@ebi.ac.uk) and Amy Curwin (amy.curwin@crg.eu)

Further details

References

This poster has been designed using images from Flaticon.com

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